Lecturer:	Date	Approved by:	Date
Final 222 CA 2 - Form 1 Phan Thi Khanh Van	*June 1st, 2023*		*June 1st, 2023*

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UNIVERSITY OF TECHNOLOGY - VNUHCM Faculty of Applied Science	**Final Exams**	Semester/Acad. year	2	2022-2023
		Date	June 12th, 2023	
	Course title	Linear Algebra		
	Course ID	MT1007	Question sheet code	1121
	Duration	100 minutes	Shift	07:00

Instructions to students:

- *This is a closed book exam. Only your calculator is allowed. Total available score: 10.*
- *At the beginning of the working time, you MUST fill in your full name and student ID on this question sheet. There are 26 questions on 4 pages. Do not round between steps. Round your final answers to 4 decimal places.*

Student's full name:	Invigilator 1: . . . . .
Student Id:	Invigilator 2: . . . . .

Part I: Multiple choice (8 points)

1. (L.O.1) Find m such that the following matrix has rank 2: $A = \begin{pmatrix} 6 & 1 & 1 \\ 0 & -5 & 2 \\ 6 & -3 & m \end{pmatrix}$.

   Ⓐ For all m. Ⓑ $m = 2.1$. Ⓒ $m = 7.6$. Ⓓ m does not exist. Ⓔ $m = 2.6$.

2. (L.O.1) Find m such that the following system only has trivial solution: $\begin{cases} 3\text{ x} + \text{y} + \text{z} = 0\ , \\ 2\text{ x -y} + 2\text{ z} = 0\ , \\ \text{-x -2 y} + \text{m z} = 0\ . \end{cases}$

   Ⓐ m is not equal to 1. Ⓑ $m = 2$. Ⓒ For all m. Ⓓ $m = 1$. Ⓔ Does not exist m.

3. (L.O.1) In $\mathbb{R}^3$ let $E = \{(1, -2, 1)^T, (0, 0, m)^T, (-2, 7, 2)^T, (10, -29, 0)^T$ be a vector set. Find all real values of m such that E is a spanning set of $\mathbb{R}^3$.

   Ⓐ m does not exist. Ⓑ For all m. Ⓒ $m = 5$. Ⓓ $m = 3$. Ⓔ m is not equal to 0.

Questions 4 through 7

A basis E for the nullspace of $A = \begin{pmatrix} -4 & 5 & 1 & 3 \\ -12 & 15 & 7 & 5 \end{pmatrix}$ has the form $\{(x, 1, y, 0), (z, 0, t, 1)\}$

4. (L.O.1) Determine x, y, z, t.

   Ⓐ $x = 2, y = -0.75, z = 1, t = 5$ Ⓑ $x = 2, y = -0.75, z = 0, t = 1.25$ Ⓒ Other answers are wrong. Ⓓ $x = 2, y = 0, z = 2, t = -5$ Ⓔ $x = 1.25, y = 0, z = 1, t = 1$

5. (L.O.2) Find u and v such that $w = (u, 0, v, 4)$ is in Null(A). Ⓐ Other answers are wrong. Ⓑ u = 4.5, v = -7.5 Ⓒ u = 4, v = 1.5 Ⓓ u = 6.5, v = -3.5 Ⓔ u = 4, v = 4

6. (L.O.1) Let θ be a vector in Null(A) such that $[\theta]_E = (-3, 2)^T$. Find the vector θ.

   Ⓐ $(-1.75, -3, 2, 2)$ Ⓑ $(-3.5, 2, 2.25, -3)$ Ⓒ $(-12, 15, 11, 18)$ Ⓓ Other answers are wrong. Ⓔ $(28, -35, -19, -37)$

7. (L.O.1) Let F be another basis of Null(A) and the transition matrix from F to E is $T_{F\to E} = [[f_1]_E[f_2]_E] = \begin{pmatrix} -3 & 2 \\ 1 & -1 \end{pmatrix}$. Find the coordinate vector of θ with respect to F.

   Ⓐ $(-0.5, -3.5)^T$. Ⓑ $(13, -5)^T$. Ⓒ $(13.5, -5.5)^T$. Ⓓ $(-1, -3)^T$. Ⓔ Other answers are wrong.

8. (L.O.1) Let u and v be two vectors in the inner product space $(V, \langle ., . \rangle)$. Assume that $\|u\| = 4$, $\|v\| = 6$ and the angle between u and v is $\pi/6$. Find the norm of the vector $\| - 2u + 2v\|$.

   Ⓐ 11.2. Ⓑ 6.4593. Ⓒ All other answers are wrong. Ⓓ 10. Ⓔ 12.8616.

Question 9 through 11:

In $\mathbb{R}^3$ we define the weighted inner product $\langle ., . \rangle$ that is given by the formula

$$\langle x, y \rangle = 6x_1y_1 + 2x_2y_2 + x_3y_3.$$

Let X be the subspace of $\mathbb{R}_3$ that is spanned by the basis $S = \{u = (-1, 0, -4), v = (-3, 2, -22)\}$.‘

9. (L.O.2) One basis of the orthogonal complement of X in $\mathbb{R}_3$ is

   Ⓐ Other answers are wrong Ⓑ (-0.6667, 2.5, 1) Ⓒ (2.5, -0.6667, 1) Ⓓ $\{(-1, 0, -4), (-3, 2, -22)\}$ Ⓔ $\{(1, 0, 4), (0, 1, -5)\}$

10. (L.O.1) The projection of $z = (1, 3, -3)$ onto $X^\perp$ is

    Ⓐ (-0.6667, 2.5, 1) Ⓑ (1.3299, 1.7629, -3.4948) Ⓒ (0.3299, -1.2371, -0.4948) Ⓓ (1, 3, -3) Ⓔ Other answers are wrong

11. (L.O.1) The distance from z to X is

    Ⓐ 1.9897 Ⓑ Other answers are wrong Ⓒ 3.9794 Ⓓ 0.4948 Ⓔ 0.9897

Question 12 through 13:

Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be a linear transformation, x and y be two vectors in $\mathbb{R}^4$ such that $T(x) = (8, 3, -1)^T$ and $T(y) = (4, 9, -4)^T$.

12. (L.O.1) Find the first component of the vector $T(-2x + y)$.

    Ⓐ 6.5. Ⓑ 6. Ⓒ -12. Ⓓ 1.5. Ⓔ All other answers are wrong.

13. (L.O.1) Let $A = \begin{bmatrix} 3 & 12 & 22 & -44 \\ 1 & 4 & 5 & -10 \\ 1 & 4 & 9 & -18 \end{bmatrix}$ be the standard transformation matrix of T. Which of the following sets is a basis of the kernel of T?

    Ⓐ $\{(1, 4, 0, 0), (0, 0, 1, -2)\}$. Ⓑ $\{(3, 1, 1), (22, 5, 9), (-44, -10, -18)\}$. Ⓒ $\{(-4, 1, 0, 0), (0, 0, 2, 1)\}$. Ⓓ $\{(3, 1, 1), (22, 5, 9)\}$. Ⓔ All other answers are wrong.

Question 14 through 17:

In $\mathbb{R}^2$ with the standard inner product let d be the line x+y = 0.

14. (L.O.1) Find the matrix of the projection transformation f about the line d.

    Ⓐ $\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$ Ⓑ $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ Ⓒ $\begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}$ Ⓓ $\begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{pmatrix}$ Ⓔ Other answers are wrong.

15. (L.O.1) Find the projection of the triangle with vertices $A(5,0), B(7,-4), C(2,-1)$ onto the line d.

Ⓐ All other answers are wrong. Ⓑ $A'(1.5,-1.5), B'(2.5,-2.5), C'(5.5,-5.5)$.
Ⓒ $A'(2.5,-2.5), B'(5.5,-5.5), C'(1.5,-1.5)$. Ⓓ $A'(-5.5,5.5), B'(1.5,-1.5), C'(2.5,-2.5)$.
Ⓔ $A'(5.5,-5.5), B'(1.5,-1.5), C'(2.5,-2.5)$.

16. (L.O.1)

Find the matrix of the rotation transformation g about the origin by $\pi/4$ counterclockwise.

Ⓐ All other answers are wrong. Ⓑ $\begin{pmatrix} 0.7071 & -0.7071 \\ 0.7071 & 0.7071 \end{pmatrix}$.
Ⓒ $\begin{pmatrix} 0.7071 & 0.7071 \\ 0.7071 & 0.7071 \end{pmatrix}$. Ⓓ $\begin{pmatrix} -1.7929 & 0.7071 \\ -0.2071 & -1.7929 \end{pmatrix}$.
Ⓔ $\begin{pmatrix} -1.7929 & -0.2071 \\ 1.2071 & -1.7929 \end{pmatrix}$.

17. (L.O.1)

Find the matrix of the composite transformation $f \circ g$.

Ⓐ $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Ⓑ $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$. Ⓒ $\begin{pmatrix} 0 & -0.7071 \\ 0 & 0.7071 \end{pmatrix}$. Ⓓ $\begin{pmatrix} 3 & 3 \\ -0.7071 & 0.7071 \end{pmatrix}$. Ⓔ All other answers are wrong.

18. (L.O.1)

Let $E = \{(-8,2)^T, (-49,14)^T\}$ and $F = \{(-8,-10)^T, (-35,-42)^T\}$ be two bases of $\mathbb{R}^2$ and $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation defined by $T(x,y) = (5x-4y, -x-2y)$. Find the matrix A_{EF} of T with respect to the bases E and F.

Ⓐ $\begin{pmatrix} 76 & 17 \\ -26 & -6 \end{pmatrix}$ Ⓑ $\begin{pmatrix} -8 & -49 \\ 2 & 14 \end{pmatrix}$ Ⓒ $\begin{pmatrix} -154 & -955.5 \\ 36.5714 & 227 \end{pmatrix}$ Ⓓ Other answers are wrong.
Ⓔ $\begin{pmatrix} -8 & -35 \\ -10 & -42 \end{pmatrix}$

19. (L.O.1) Let $A = \begin{pmatrix} 2 & -5 \\ -20 & 17 \end{pmatrix}$. Which of the following vectors in an eigenvector of A?

Ⓐ $(5,-4)$ Ⓑ All other answers are wrong. Ⓒ $(-5,-3)$ Ⓓ $(-5,-5)$ Ⓔ $(5,-3)$

Questions 20 through 22 Suppose that $u = (5,1)^T$ is an eigenvector of the matrix A corresponding to $\lambda = -3$, and $v = (4,0)^T$ is an eigenvector of A corresponding to $\lambda = 1$.

20. (L.O.1) Find $det(A)$.

Ⓐ All other answers are wrong. Ⓑ -3 Ⓒ 0 Ⓓ -5 Ⓔ -6

21. (L.O.1) Find the vector $A \times (3u + 2v)$.

Ⓐ All other answers are wrong. Ⓑ (-6, -39) Ⓒ (-9, -37) Ⓓ (-37, -9) Ⓔ (-39, -6)

22. (L.O.1) Find the $(1,2)$ entry of the matrix A.

Ⓐ -23.5 Ⓑ -20 Ⓒ All other answers are wrong. Ⓓ -19.5 Ⓔ -16.5

23. (L.O.1) Let $f : \mathbb{R}^n \to \mathbb{R}^n$ and $g : \mathbb{R}^n \to \mathbb{R}^n$ be two linear transformations. Choose the WRONG statement.

Ⓐ For all $\alpha \in \mathbb{R}$: $h(x) := \alpha f(x)$ is a linear transformation. Ⓑ All other answers are wrong.
Ⓒ For all $\alpha, \beta \in \mathbb{R}$: $h(x) := \alpha f(x) + \beta g(x)$ is a linear transformation. Ⓓ If $n = 1$, then the function $h(x) := \frac{f(x)}{g(x)}$ is a linear transformation. Ⓔ $h(x) := f(x) + g(x)$ is a linear transformation.

Question 24 through 25:

A company took out a loan of 856 (thoudsand dollars) to launch a business project. The loan was divided into three parts, with some borrowed at 7(%), some at 9(%), and some at 12(%). To pay the annual interest of 76.4 (thoundsand dollars), the business needs to determine how much was borrowed at each rate. Additionally, it's known that the amount borrowed at 9(%) is four times the amount borrowed at 12(%).

24. (L.O.1) Find the amount that the company borrowed at the rate of 7 (%) Ⓐ 222.6638 Ⓑ 222.1538 Ⓒ 222.6438 Ⓓ 222.5138 Ⓔ All other answers are wrong.

25. (L.O.1) Find the amount that the company borrowed at the rate of 9 (%) Ⓐ 507.4369 Ⓑ 507.5669 Ⓒ All other answers are wrong. Ⓓ 507.5869 Ⓔ 507.0769

Part II: Essay (2 points)

26. (L.O.1) The rescue force in a densely populated city in the United States consisted of 500 vehicles distributed in three areas: the airport, the East office, and the West office. Each day after the rescue vehicle completes its mission, it will return to one of the three locations above, the closest place to it. According to long-term observation, after each day 15% of rescue force that start from the airport will return to the East office and 5% will return to the West office; Of the force that start from the East office, 5% will return to the Airport and 5% will return to the West office; Of the force that start from the West office, 10% will return to the airport and 5% will return to the East office.

    a. What is the transition matrix? (The 3 states should be in this order: Airport, East Office, West Office)

    b. What are the eigenvalues and corresponding eigenvectors of the transition matrix?

    c. Find the stable distribution vector (the distribution vector that won't change after every day).

    d. Given that today there are $200, 200, 100$ vehicles start from the airport, East Office, West Office, respectively. Using the diagonalization method to find the number of vehicles in each class after 100 days.

Lecturer:	Date	Approved by:	Date
Final 222 CA 2 - Form 2 Phan Thi Khanh Van	*June 1st, 2023*		*June 1st, 2023*

UNIVERSITY OF TECHNOLOGY - VNUHCM Faculty of Applied Science	**Final Exams**	Semester/Acad. year	2	2022-2023
		Date	June 12th, 2023	
	Course title	Linear Algebra		
	Course ID	MT1007	Question sheet code	1122
	Duration	100 minutes	Shift	07:00

Instructions to students:

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- *At the beginning of the working time, you MUST fill in your full name and student ID on this question sheet. There are 27 questions on 4 pages. Do not round between steps. Round your final answers to 4 decimal places.*

Student's full name:	Invigilator 1: .....
Student Id:	Invigilator 2: .....

Part I: Multiple choice (8 points)

1. (L.O.1) Find m such that the following matrix has rank 3: $A = \begin{pmatrix} 0 & 5 & 1 \\ 6 & -2 & 2 \\ -5 & -4 & m \end{pmatrix}$.

   Ⓐ For all m. Ⓑ m does not exist. Ⓒ $m = 0.7$. Ⓓ m is not equal to -2.8. Ⓔ $m = -4.3$.

2. (L.O.1) Find m such that the following system only has trivial solution: $\begin{cases} \text{-3 x + y + z} = 0\ , \\ \text{7 x -y + 2 z} = 0\ , \\ \text{-5 x + 6 y + m z} = 0\ . \end{cases}$

   Ⓐ m is not equal to 15.75. Ⓑ For all m. Ⓒ Does not exist m. Ⓓ $m = 15.75$. Ⓔ $m = 15.25$.

3. (L.O.1) In $\mathbb{R}^3$ let $E = \{(1,1,1)^T, (7,3,m)^T, (-4,7,2)^T, (-2,30,-22)^T$ be a vector set. Find all real values of m such that E is a spanning set of $\mathbb{R}^3$.

   Ⓐ m does not exist. Ⓑ $m = 0.3182$. Ⓒ $m = 9.8182$. Ⓓ For all m. Ⓔ m is not equal to 4.8182.

Questions 4 through 7

A basis E for the nullspace of $A = \begin{pmatrix} -4 & -3 & 2 & -6 \\ -8 & -6 & 7 & -15 \end{pmatrix}$ has the form $\{(x,1,y,0),(z,0,t,1)\}$

4. (L.O.1) Determine x, y, z, t.

   Ⓐ $x = -1, y = 0, z = -1, t = -5$ Ⓑ $x = -1, y = -0.3333, z = 0, t = -0.75$ Ⓒ $x = -1, y = -0.3333, z = 1, t = 5$ Ⓓ Other answers are wrong. Ⓔ $x = -0.75, y = 0, z = -1, t = 1$

5. (L.O.2) Find u and v such that $w = (u, 3, v, -2)$ is in Null(A). Ⓐ u = -0.25, v = -2 Ⓑ u = 2.25, v = 0.1666 Ⓒ u = 4.25, v = 2.6666 Ⓓ u = -2.75, v = 2.1666 Ⓔ Other answers are wrong.

6. (L.O.1) Let θ be a vector in Null(A) such that $[\theta]_E = (-3,4)^T$. Find the vector θ.

   Ⓐ $(-1.75, -3, 4, 4)$ Ⓑ Other answers are wrong. Ⓒ $(0, 4, 0.9999, -3)$ Ⓓ $(8, 6, -13, 7)$ Ⓔ $(-20, -15, 22, -21)$

7. (L.O.1) Let F be another basis of Null(A) and the transition matrix from F to E is $T_{F\to E} = [[f_1]_E[f_2]_E] = \begin{pmatrix} -7 & 10 \\ 16 & -23 \end{pmatrix}$. Find the coordinate vector of θ with respect to F.

   Ⓐ $(29.5, 19.5)^T$. Ⓑ $(29, 20)^T$. Ⓒ Other answers are wrong. Ⓓ $(61.5, -140.5)^T$. Ⓔ $(61, -140)^T$.

8. (L.O.1) Let u and v be two vectors in the inner product space $(V, \langle ., . \rangle)$. Assume that $\|u\| = 6$, $\|v\| = 2$ and the angle between u and v is $\pi/6$. Find the norm of the vector $\|2u - 4v\|$.

   Ⓐ All other answers are wrong. Ⓑ 16. Ⓒ 6.4593. Ⓓ 14.8616. Ⓔ 11.2.

Question 9 through 11:

In $\mathbb{R}^3$ we define the weighted inner product $\langle ., . \rangle$ that is given by the formula

$$\langle x, y \rangle = x_1y_1 + 6x_2y_2 + 5x_3y_3.$$

Let X be the subspace of $\mathbb{R}_3$ that is spanned by the basis $S = \{u = (4, -3, 31), v = (0, 1, -5)\}$.‘

9. (L.O.2) One basis of the orthogonal complement of X in $\mathbb{R}_3$ is

   Ⓐ $\{(4, -3, 31), (0, 1, -5)\}$ Ⓑ Other answers are wrong Ⓒ (-20, 4.1667, 1) Ⓓ (4.1667, -20, 1) Ⓔ $\{(1, 0, 4), (0, 1, -5)\}$

10. (L.O.1) The projection of $z = (3, 2, 2)$ onto $X^\perp$ is

    Ⓐ (0, -0, -0) Ⓑ (3, 2, 2) Ⓒ Other answers are wrong Ⓓ (-20, 4.1667, 1) Ⓔ (3, 2, 2)

11. (L.O.1) The distance from z to X is

    Ⓐ 0 Ⓑ Other answers are wrong Ⓒ 0 Ⓓ 0 Ⓔ 0

Question 12 through 13:

Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be a linear transformation, x and y be two vectors in $\mathbb{R}^4$ such that $T(x) = (2, 4, -1)^T$ and $T(y) = (-8, 5, 1)^T$.

12. (L.O.1) Find the second component of the vector $T(3x + y)$.

    Ⓐ All other answers are wrong. Ⓑ 15.5. Ⓒ 17. Ⓓ 22. Ⓔ 21.

13. (L.O.1) Let $A = \begin{bmatrix} -1 & -3 & -4 & 20 \\ 1 & 3 & 2 & -10 \\ 6 & 18 & 13 & -65 \end{bmatrix}$ be the standard transformation matrix of T. Which of the following sets is a basis of the kernel of T?

    Ⓐ $\{(-3, 1, 0, 0), (0, 0, 5, 1)\}$. Ⓑ $\{(-1, 1, 6), (-4, 2, 13), (20, -10, -65)\}$. Ⓒ $\{(1, 3, 0, 0), (0, 0, 1, -5)\}$. Ⓓ $\{(-1, 1, 6), (-4, 2, 13)\}$. Ⓔ All other answers are wrong.

Question 14 through 17:

In $\mathbb{R}^2$ with the standard inner product let d be the line x-2y = 0.

14. (L.O.1) Find the matrix of the projection transformation f about the line d.

    Ⓐ $\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$ Ⓑ $\begin{pmatrix} 0.8 & 0.4 \\ 0.4 & 0.2 \end{pmatrix}$ Ⓒ Other answers are wrong. Ⓓ $\begin{pmatrix} 0.8 & -0.4 \\ -0.4 & 0.2 \end{pmatrix}$ Ⓔ $\begin{pmatrix} 0.6 & 0.8 \\ 0.8 & -0.6 \end{pmatrix}$

15. (L.O.1) Find the projection of the triangle with vertices $A(-3,1), B(-5,5), C(-4,3)$ onto the line d.

Ⓐ $A'(-2,-1), B'(-2,-1), C'(-2,-1)$. Ⓑ $A'(-2,-1), B'(-2,-1), C'(-2,-1)$. Ⓒ $A'(-2,-1), B'($
Ⓓ $A'(-1,-2), B'(-2,-1), C'(-2,-1)$. Ⓔ All other answers are wrong.

16. (L.O.1)

Find the matrix of the rotation transformation g about the origin by $\pi/3$ counterclockwise.

Ⓐ $\begin{pmatrix} 0.5 & 0.866 \\ 0.5 & 0.5 \end{pmatrix}$. Ⓑ All other answers are wrong.
Ⓒ $\begin{pmatrix} 1 & 0.866 \\ 2.134 & 3.5 \end{pmatrix}$. Ⓓ $\begin{pmatrix} 1 & 2.134 \\ 2.366 & 3.5 \end{pmatrix}$.
Ⓔ $\begin{pmatrix} 0.5 & -0.866 \\ 0.866 & 0.5 \end{pmatrix}$.

17. (L.O.1)

Find the matrix of the composite transformation $g \circ f$.

Ⓐ $\begin{pmatrix} 2.0536 & 0.8928 \\ 1.0268 & 2.4464 \end{pmatrix}$. Ⓑ $\begin{pmatrix} 0.0536 & 0.0268 \\ 0.8928 & 0.4464 \end{pmatrix}$. Ⓒ $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Ⓓ All other answers are wrong.
Ⓔ $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$.

18. (L.O.1)

Let $E = \{(3,-8)^T, (-1,5)^T\}$ and $F = \{(-5,2)^T, (4,-1)^T\}$ be two bases of $\mathbb{R}^2$ and $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation defined by $T(x,y) = (3x+4y, x+2y)$. Find the matrix A_{EF} of T with respect to the bases E and F.

Ⓐ $\begin{pmatrix} 31 & -18 \\ -86 & 50 \end{pmatrix}$ Ⓑ Other answers are wrong. Ⓒ $\begin{pmatrix} -60 & -22 \\ 16 & 6 \end{pmatrix}$ Ⓓ $\begin{pmatrix} -25 & 17.6667 \\ -37 & 26.3333 \end{pmatrix}$
Ⓔ $\begin{pmatrix} 63 & -49 \\ -33 & 25 \end{pmatrix}$

19. (L.O.1) Let $A = \begin{pmatrix} 7 & -2 \\ -16 & 11 \end{pmatrix}$. Which of the following vectors in an eigenvector of A?

Ⓐ All other answers are wrong. Ⓑ $(-2,-4)$ Ⓒ $(4,-4)$ Ⓓ $(4,3)$ Ⓔ $(-2,3)$

Questions 20 through 22 Suppose that $u = (5,-3)^T$ is an eigenvector of the matrix A corresponding to $\lambda = -2$, and $v = (1,-1)^T$ is an eigenvector of A corresponding to $\lambda = -3$.

20. (L.O.1) Find $det(A)$.

Ⓐ All other answers are wrong. Ⓑ 11 Ⓒ 6 Ⓓ 5 Ⓔ 9

21. (L.O.1) Find the vector $A \times (-3u + 2v)$.

Ⓐ All other answers are wrong. Ⓑ (21, -14) Ⓒ (24, -12) Ⓓ (-14, 21) Ⓔ (-12, 24)

22. (L.O.1) Find the $(1,2)$ entry of the matrix A.

Ⓐ All other answers are wrong. Ⓑ 5 Ⓒ 0 Ⓓ 7.5 Ⓔ 2.5

23. (L.O.1) Let $f : \mathbb{R}^n \to \mathbb{R}^n$ and $g : \mathbb{R}^n \to \mathbb{R}^n$ be two linear transformations. Choose the WRONG statement.

Ⓐ $h(x) := f(x) + g(x)$ is a linear transformation. Ⓑ If $n = 1$, then the function $f^g(x) := (f(x))^{g(x)}$ is a linear transformation Ⓒ The composite transformation $g \circ f(x) := g(f(x))$ is a

linear transformation. Ⓓ For all $\alpha \in \mathbb{R}$: $h(x) := \alpha f(x)$ is a linear transformation. Ⓔ All other answers are wrong.

Question 24 through 25:

A company took out a loan of 864 (thoudsand dollars) to launch a business project. The loan was divided into three parts, with some borrowed at 7(%), some at 9(%), and some at 12(%). To pay the annual interest of 72.8 (thoundsand dollars), the business needs to determine how much was borrowed at each rate. Additionally, it's known that the amount borrowed at 9(%) is four times the amount borrowed at 12(%).

24. (L.O.1) Find the amount that the company borrowed at the rate of 7 (%) Ⓐ 390.6038 Ⓑ 391.1038 Ⓒ 390.1538 Ⓓ All other answers are wrong. Ⓔ 390.4338

25. (L.O.1) Find the amount that the company borrowed at the rate of 9 (%) Ⓐ All other answers are wrong. Ⓑ 379.3569 Ⓒ 379.5269 Ⓓ 380.0269 Ⓔ 379.0769

Part II: Essay (2 points)

26. (L.O.1)

27. (L.O.1) The rescue force in a densely populated city in the United States consisted of 500 vehicles distributed in three areas: the airport, the East office, and the West office. Each day after the rescue vehicle completes its mission, it will return to one of the three locations above, the closest place to it. According to long-term observation, after each day 15% of rescue force that start from the airport will return to the East office and 5% will return to the West office; Of the force that start from the East office, 5% will return to the Airport and 5% will return to the West office; Of the force that start from the West office, 10% will return to the airport and 5% will return to the East office.

    a. What is the transition matrix? (The 3 states should be in this order: Airport, East Office, West Office)

    b. What are the eigenvalues and corresponding eigenvectors of the transition matrix?

    c. Find the stable distribution vector (the distribution vector that won't change after every day).

    d. Given that today there are $200, 200, 100$ vehicles start from the airport, East Office, West Office, respectively. Using the diagonalization method to find the number of vehicles in each class after 100 days.

Lecturer:	Date	Approved by:	Date
Final 222 CA 2 - Form 3 Phan Thi Khanh Van	*June 1st, 2023*		*June 1st, 2023*

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UNIVERSITY OF TECHNOLOGY - VNUHCM Faculty of Applied Science	**Final Exams**	Semester/Acad. year	2	2022-2023
		Date	June 12th, 2023	
	Course title	Linear Algebra		
	Course ID	MT1007	Question sheet code	1123
	Duration	100 minutes	Shift	07:00

Instructions to students:

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- *At the beginning of the working time, you MUST fill in your full name and student ID on this question sheet. There are 26 questions on 4 pages. Do not round between steps. Round your final answers to 4 decimal places.*

Student's full name:	Invigilator 1: . . . . .
Student Id:	Invigilator 2: . . . . .

Part I: Multiple choice (8 points)

1. (L.O.1) Find m such that the following matrix has rank 3: $A = \begin{pmatrix} 2 & -4 & 1 \\ 0 & -5 & 2 \\ -5 & 6 & m \end{pmatrix}$.

   Ⓐ $m = 4.1$. Ⓑ m is not equal to -0.9. Ⓒ $m = -3.4$. Ⓓ For all m. Ⓔ m does not exist.

2. (L.O.1) Find m such that the following system only has trivial solution: $\begin{cases} \text{-2 y + z = 0 ,} \\ \text{-5 x -2 y + 2 z = 0 ,} \\ \text{5 x -y + m z = 0 .} \end{cases}$

   Ⓐ Does not exist m. Ⓑ m is not equal to -0.5. Ⓒ For all m. Ⓓ $m = -0.5$. Ⓔ $m = -5.5$.

3. (L.O.1) In $\mathbb{R}^3$ let $E = \{(6, -5, 1)^T, (-2, 2, m)^T, (1, 7, 2)^T, (-23, 4, 8)^T$ be a vector set. Find all real values of m such that E is not a spanning set of $\mathbb{R}^3$.

   Ⓐ $m = 2.2447$. Ⓑ For all m. Ⓒ m does not exist. Ⓓ $m = -0.7553$. Ⓔ $m = -0.2553$.

Questions 4 through 7

A basis E for the nullspace of $A = \begin{pmatrix} -3 & 4 & -2 & -5 \\ -9 & 12 & -1 & 10 \end{pmatrix}$ has the form $\{(x, 1, y, 0), (z, 0, t, 1)\}$

4. (L.O.1) Determine x, y, z, t.

   Ⓐ $x = 1.3333, y = 0, z = -1, t = -1$ Ⓑ $x = -1, y = 0, z = -1, t = -5$ Ⓒ Other answers are wrong. Ⓓ $x = -1, y = -0.4, z = 1, t = 5$ Ⓔ $x = -1, y = -0.4, z = 0, t = 1.3333$

5. (L.O.2) Find u and v such that $w = (u, 0, v, 0)$ is in Null(A). Ⓐ u = 0.5, v = -3 Ⓑ u = 0, v = 0 Ⓒ u = -3.5, v = -1 Ⓓ Other answers are wrong. Ⓔ u = -0.5, v = -4.5

6. (L.O.1) Let θ be a vector in Null(A) such that $[\theta]_E = (5, 3)^T$. Find the vector θ.

   Ⓐ $(-54, 72, -11, -8)$ Ⓑ Other answers are wrong. Ⓒ $(-42, 56, -13, -8)$ Ⓓ $(3.6667, 5, -3, 3)$ Ⓔ $(-1.0001, 3, -2, 5)$

7. (L.O.1) Let F be another basis of Null(A) and the transition matrix from F to E is $T_{F\to E} = [[f_1]_E[f_2]_E] = \begin{pmatrix} -1 & 1 \\ -2 & 1 \end{pmatrix}$. Find the coordinate vector of θ with respect to F.

   Ⓐ $(-1.5, -7.5)^T$. Ⓑ $(-2, -7)^T$. Ⓒ $(2, 7)^T$. Ⓓ $(2.5, 6.5)^T$. Ⓔ Other answers are wrong.

8. (L.O.1) Let u and v be two vectors in the inner product space $(V, \langle ., . \rangle)$. Assume that $\|u\| = 6$, $\|v\| = 2$ and the angle between u and v is $5\pi/6$. Find the norm of the vector $\| - 4u - 3v\|$.

   Ⓐ All other answers are wrong. Ⓑ 75.6. Ⓒ 19.0417. Ⓓ 68.4. Ⓔ 48.7292.

Question 9 through 11:

In $\mathbb{R}^3$ we define the weighted inner product $\langle ., . \rangle$ that is given by the formula

$$\langle x, y \rangle = 6x_1y_1 + 2x_2y_2 + 7x_3y_3.$$

Let X be the subspace of $\mathbb{R}_3$ that is spanned by the basis $S = \{u = (-2, 0, -8), v = (-2, -2, 2)\}$.‘

9. (L.O.2) One basis of the orthogonal complement of X in $\mathbb{R}_3$ is

   Ⓐ $\{(-2, 0, -8), (-2, -2, 2)\}$ Ⓑ (-4.6667, 17.5, 1) Ⓒ Other answers are wrong Ⓓ (17.5, -4.6667, 1) Ⓔ $\{(1, 0, 4), (0, 1, -5)\}$

10. (L.O.1) The projection of $z = (-4, -1, 3)$ onto $X^\perp$ is

    Ⓐ (-4.6667, 17.5, 1) Ⓑ (-4, -1, 3) Ⓒ (-3.3904, -3.2862, 2.8694) Ⓓ (0.6096, -2.2862, -0.1306) Ⓔ Other answers are wrong

11. (L.O.1) The distance from z to X is

    Ⓐ 7.1562 Ⓑ 0.2613 Ⓒ Other answers are wrong Ⓓ 0.1306 Ⓔ 3.5781

Question 12 through 13:

Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be a linear transformation, x and y be two vectors in $\mathbb{R}^4$ such that $T(x) = (-10, -2, 1)^T$ and $T(y) = (2, -8, 4)^T$.

12. (L.O.1) Find the second component of the vector $T(-3x + y)$.

    Ⓐ 18.5. Ⓑ All other answers are wrong. Ⓒ 25. Ⓓ 32. Ⓔ 21.

13. (L.O.1) Let $A = \begin{bmatrix} -2 & -8 & 11 & -22 \\ -1 & -4 & 7 & -14 \\ 0 & 0 & 1 & -2 \end{bmatrix}$ be the standard transformation matrix of T. Which of the following sets is a basis of the kernel of T?

    Ⓐ $\{(-2, -1, 0), (11, 7, 1)\}$. Ⓑ $\{(-2, -1, 0), (11, 7, 1), (-22, -14, -2)\}$. Ⓒ $\{(1, 4, 0, 0), (0, 0, 1, -2)\}$. Ⓓ All other answers are wrong. Ⓔ $\{(-4, 1, 0, 0), (0, 0, 2, 1)\}$.

Question 14 through 17:

In $\mathbb{R}^2$ with the standard inner product let d be the line x+2y = 0.

14. (L.O.1) Find the matrix of the projection transformation f about the line d.

    Ⓐ $\begin{pmatrix} 0.8 & 0.4 \\ 0.4 & 0.2 \end{pmatrix}$ Ⓑ Other answers are wrong. Ⓒ $\begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix}$ Ⓓ $\begin{pmatrix} 0.6 & -0.8 \\ -0.8 & -0.6 \end{pmatrix}$ Ⓔ $\begin{pmatrix} 0.8 & -0.4 \\ -0.4 & 0.2 \end{pmatrix}$

15. (L.O.1) Find the projection of the triangle with vertices $A(0,-4), B(6,-1), C(5,1)$ onto the line d.

Ⓐ $A'(1.6,-0.8), B'(5.2,-2.6), C'(3.6,-1.8)$. Ⓑ $A'(3.6,-1.8), B'(1.6,-0.8), C'(5.2,-2.6)$.
Ⓒ $A'(5.2,-2.6), B'(3.6,-1.8), C'(1.6,-0.8)$. Ⓓ $A'(-2.6,5.2), B'(3.6,-1.8), C'(1.6,-0.8)$.
Ⓔ All other answers are wrong.

16. (L.O.1)

Find the matrix of the rotation transformation g about the origin by $\pi/3$ clockwise.

Ⓐ $\begin{pmatrix} 0.5 & -0.866 \\ 0.5 & 0.5 \end{pmatrix}$. Ⓑ $\begin{pmatrix} 0 & -0.866 \\ -1.134 & -2.5 \end{pmatrix}$.
Ⓒ $\begin{pmatrix} 0 & -1.134 \\ 1.134 & -2.5 \end{pmatrix}$. Ⓓ All other answers are wrong.
Ⓔ $\begin{pmatrix} 0.5 & 0.866 \\ -0.866 & 0.5 \end{pmatrix}$.

17. (L.O.1)

Find the matrix of the composite transformation $g \circ f$.

Ⓐ $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$. Ⓑ $\begin{pmatrix} 0.0536 & -0.0268 \\ -0.8928 & 0.4464 \end{pmatrix}$. Ⓒ All other answers are wrong. Ⓓ $\begin{pmatrix} 0.0536 & 2.1072 \\ 0.9732 & 2.4464 \end{pmatrix}$.
Ⓔ $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

18. (L.O.1)

Let $E = \{(1,-1)^T, (-5,7)^T\}$ and $F = \{(-1,3)^T, (5,-3)^T\}$ be two bases of $\mathbb{R}^2$ and $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation defined by $T(x,y) = (-4x+y, -3x+3y)$. Find the matrix A_{EF} of T with respect to the bases E and F.

Ⓐ Other answers are wrong. Ⓑ $\begin{pmatrix} -3.75 & 21.75 \\ -1.75 & 9.75 \end{pmatrix}$ Ⓒ $\begin{pmatrix} -25 & 153 \\ 3 & -27 \end{pmatrix}$ Ⓓ $\begin{pmatrix} 140 & -53 \\ -193 & 73 \end{pmatrix}$
Ⓔ $\begin{pmatrix} -28 & -11 \\ 17 & 7 \end{pmatrix}$

19. (L.O.1) Let $A = \begin{pmatrix} 4 & -1 \\ 15 & -4 \end{pmatrix}$. Which of the following vectors in an eigenvector of A?

Ⓐ $(3,5)$ Ⓑ $(-1,1)$ Ⓒ All other answers are wrong. Ⓓ $(-1,-3)$ Ⓔ $(3,1)$

Questions 20 through 22 Suppose that $u = (3,1)^T$ is an eigenvector of the matrix A corresponding to $\lambda = -5$, and $v = (-5,2)^T$ is an eigenvector of A corresponding to $\lambda = 5$.

20. (L.O.1) Find $det(A)$.

Ⓐ -30 Ⓑ -23 Ⓒ All other answers are wrong. Ⓓ -25 Ⓔ -27

21. (L.O.1) Find the vector $A \times (3u - v)$.

Ⓐ (-20, -25) Ⓑ (-25, -20) Ⓒ All other answers are wrong. Ⓓ (-22, -23) Ⓔ (-23, -22)

22. (L.O.1) Find the $(1,2)$ entry of the matrix A.

Ⓐ -10.6364 Ⓑ All other answers are wrong. Ⓒ -18.1364 Ⓓ -12.6364 Ⓔ -13.6364

23. (L.O.1) Let $f : \mathbb{R}^n \to \mathbb{R}^n$ and $g : \mathbb{R}^n \to \mathbb{R}^n$ be two linear transformations. Choose the WRONG statement.

Ⓐ The composite transformation $g \circ f(x) := g(f(x))$ is a linear transformation. Ⓑ If $n = 1$, then the function $h(x) := \frac{f(x)}{g(x)}$ is a linear transformation. Ⓒ All other answers are wrong.

Ⓓ For all $\alpha, \beta \in \mathbb{R}$: $h(x) := \alpha f(x) + \beta g(x)$ is a linear transformation. Ⓔ $h(x) := f(x) + g(x)$ is a linear transformation.

Question 24 through 25:

A company took out a loan of 831 (thoudsand dollars) to launch a business project. The loan was divided into three parts, with some borrowed at 8(%), some at 10(%), and some at 12(%). To pay the annual interest of 70.8 (thoundsand dollars), the business needs to determine how much was borrowed at each rate. Additionally, it's known that the amount borrowed at 10(%) is four times the amount borrowed at 12(%).

24. (L.O.1) Find the amount that the company borrowed at the rate of 8 (%) Ⓐ 651 Ⓑ 651.06 Ⓒ 651.99 Ⓓ 651.19 Ⓔ All other answers are wrong.

25. (L.O.1) Find the amount that the company borrowed at the rate of 10 (%) Ⓐ All other answers are wrong. Ⓑ 144.99 Ⓒ 144.19 Ⓓ 144 Ⓔ 144.06

Part II: Essay (2 points)

26. (L.O.1) The rescue force in a densely populated city in the United States consisted of 500 vehicles distributed in three areas: the airport, the East office, and the West office. Each day after the rescue vehicle completes its mission, it will return to one of the three locations above, the closest place to it. According to long-term observation, after each day 15% of rescue force that start from the airport will return to the East office and 5% will return to the West office; Of the force that start from the East office, 5% will return to the Airport and 5% will return to the West office; Of the force that start from the West office, 10% will return to the airport and 5% will return to the East office.

    a. What is the transition matrix? (The 3 states should be in this order: Airport, East Office, West Office)

    b. What are the eigenvalues and corresponding eigenvectors of the transition matrix?

    c. Find the stable distribution vector (the distribution vector that won't change after every day).

    d. Given that today there are $200, 200, 100$ vehicles start from the airport, East Office, West Office, respectively. Using the diagonalization method to find the number of vehicles in each class after 100 days.

Lecturer:	Date	Approved by:	Date
Final 222 CA 2 - Form 4 Phan Thi Khanh Van	*June 1st, 2023*		*June 1st, 2023*

UNIVERSITY OF TECHNOLOGY - VNUHCM Faculty of Applied Science	**Final Exams**	Semester/Acad. year	2	2022-2023
		Date	June 12th, 2023	
	Course title	Linear Algebra		
	Course ID	MT1007	Question sheet code	1124
	Duration	100 minutes	Shift	07:00

Instructions to students:

- *This is a closed book exam. Only your calculator is allowed. Total available score: 10.*
- *At the beginning of the working time, you MUST fill in your full name and student ID on this question sheet. There are 26 questions on 4 pages. Do not round between steps. Round your final answers to 4 decimal places.*

Student's full name:	Invigilator 1: .....
Student Id:	Invigilator 2: .....

Part I: Multiple choice (8 points)

1. (L.O.1) Find m such that the following matrix has rank 3: $A = \begin{pmatrix} 5 & -2 & 1 \\ -3 & 0 & 2 \\ 4 & 7 & m \end{pmatrix}$.

   Ⓐ $m = -18.8333$. Ⓑ For all m. Ⓒ $m = -14.8333$. Ⓓ m is not equal to -17.8333. Ⓔ m does not exist.

2. (L.O.1) Find m such that the following system only has trivial solution: $\begin{cases} \text{y} + \text{z} = 0\ , \\ 3\ \text{x} - 3\ \text{y} + 2\ \text{z} = 0\ , \\ 5\ \text{x} + \text{m}\ \text{z} = 0\ . \end{cases}$

   Ⓐ Does not exist m. Ⓑ $m = 7.8333$. Ⓒ m is not equal to 8.3333. Ⓓ For all m. Ⓔ $m = 8.3333$.

3. (L.O.1) In $\mathbb{R}^3$ let $E = \{(3,-3,1)^T, (3,2,m)^T, (7,2,2)^T, (18,15,3)^T$ be a vector set. Find all real values of m such that E is a spanning set of $\mathbb{R}^3$.

   Ⓐ m is not equal to 0.8148. Ⓑ $m = -0.6852$. Ⓒ m does not exist. Ⓓ For all m. Ⓔ $m = 2.8148$.

Questions 4 through 7

A basis E for the nullspace of $A = \begin{pmatrix} 5 & 1 & -2 & 8 \\ 10 & 2 & 1 & 21 \end{pmatrix}$ has the form $\{(x,1,y,0),(z,0,t,1)\}$

4. (L.O.1) Determine x, y, z, t.

   Ⓐ $x = -0.2, y = 0, z = -2, t = -1$ Ⓑ $x = -2, y = -1, z = 0, t = -0.2$ Ⓒ $x = -2, y = 0, z = -2, t = -5$ Ⓓ $x = -2, y = -1, z = 1, t = 5$ Ⓔ Other answers are wrong.

5. (L.O.2) Find u and v such that $w = (u,-1,v,1)$ is in Null(A). Ⓐ u = 1.2, v = -1.5 Ⓑ u = -2.3, v = -6 Ⓒ u = -6.3, v = -3.5 Ⓓ Other answers are wrong. Ⓔ u = -1.8, v = -1

6. (L.O.1) Let θ be a vector in Null(A) such that $[\theta]_E = (-3,1)^T$. Find the vector θ.

   Ⓐ $(5.8,1,3,-3)$ Ⓑ $(-1.4,-3,-1,1)$ Ⓒ Other answers are wrong. Ⓓ $(-25,-5,-5,-10)$ Ⓔ $(-5,-1,7,6)$

7. (L.O.1) Let F be another basis of Null(A) and the transition matrix from F to E is $T_{F\to E} = [[f_1]_E[f_2]_E] = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$. Find the coordinate vector of θ with respect to F.

   Ⓐ $(-1,1)^T$. Ⓑ Other answers are wrong. Ⓒ $(-0.5,0.5)^T$. Ⓓ $(-4.5,0.5)^T$. Ⓔ $(-5,1)^T$.

8. (L.O.1) Let u and v be two vectors in the inner product space $(V,\langle .,.\rangle)$. Assume that $\|u\| = 5$, $\|v\| = 7$ and the angle between u and v is $2\pi/3$. Find the norm of the vector $\| - 4u - 3v\|$.

   Ⓐ 12.0473. Ⓑ 63.1. Ⓒ 20.5183. Ⓓ 15.6846. Ⓔ All other answers are wrong.

Question 9 through 11:

In $\mathbb{R}^3$ we define the weighted inner product $\langle .,.\rangle$ that is given by the formula

$$\langle x,y\rangle = x_1y_1 + 7x_2y_2 + 6x_3y_3.$$

Let X be the subspace of $\mathbb{R}_3$ that is spanned by the basis $S = \{u = (1,0,4), v = (-4,-1,-11)\}$.‘

9. (L.O.2) One basis of the orthogonal complement of X in $\mathbb{R}_3$ is

   Ⓐ (-24, 4.2857, 1) Ⓑ Other answers are wrong Ⓒ (4.2857, -24, 1) Ⓓ $\{(1,0,4),(-4,-1,-11)\}$ Ⓔ $\{(1,0,4),(0,1,-5)\}$

10. (L.O.1) The projection of $z = (-5,-5,-4)$ onto $X^\perp$ is

    Ⓐ (-1.8239, 0.3257, 0.076) Ⓑ (-5, -5, -4) Ⓒ (-6.8239, -4.6743, -3.924) Ⓓ Other answers are wrong Ⓔ (-24, 4.2857, 1)

11. (L.O.1) The distance from z to X is

    Ⓐ -0.076 Ⓑ -0.152 Ⓒ Other answers are wrong Ⓓ 4.0516 Ⓔ 2.0258

Question 12 through 13:

Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be a linear transformation, x and y be two vectors in $\mathbb{R}^4$ such that $T(x) = (8,5,9)^T$ and $T(y) = (-10,10,-2)^T$.

12. (L.O.1) Find the second component of the vector $T(-x+y)$.

    Ⓐ All other answers are wrong. Ⓑ 2.5. Ⓒ 6. Ⓓ 5. Ⓔ 8.

13. (L.O.1) Let $A = \begin{bmatrix} -1 & -4 & 1 & -5 \\ -5 & -20 & 3 & -15 \\ -5 & -20 & 4 & -20 \end{bmatrix}$ be the standard transformation matrix of T. Which of the following sets is a basis of the kernel of T?

    Ⓐ $\{(-1,-5,-5),(1,3,4),(-5,-15,-20)\}$. Ⓑ All other answers are wrong. Ⓒ $\{(-1,-5,-5),(1,3,$ Ⓓ $\{(1,4,0,0),(0,0,1,-5)\}$. Ⓔ $\{(-4,1,0,0),(0,0,5,1)\}$.

Question 14 through 17:

In $\mathbb{R}^2$ with the standard inner product let d be the line x-y = 0.

14. (L.O.1) Find the matrix of the projection transformation f about the line d.

    Ⓐ $\begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}$ Ⓑ $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ Ⓒ Other answers are wrong. Ⓓ $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ Ⓔ $\begin{pmatrix} -0.5 & 0.5 \\ 0.5 & -0.5 \end{pmatrix}$

15. (L.O.1) Find the projection of the triangle with vertices $A(6,-1), B(7,-5), C(-6,2)$ onto the line d.

Ⓐ $A'(1,1), B'(-2,-2), C'(2.5,2.5)$. Ⓑ $A'(2.5,2.5), B'(1,1), C'(-2,-2)$.
Ⓒ $A'(-2,-2), B'(2.5,2.5), C'(1,1)$. Ⓓ $A'(1,1), B'(-2,-2), C'(2.5,2.5)$.
Ⓔ All other answers are wrong.

16. (L.O.1)

Find the matrix of the rotation transformation g about the origin by $\pi/2$ clockwise.

Ⓐ All other answers are wrong. Ⓑ $\begin{pmatrix} -1 & -1 \\ -1.5 & -1 \end{pmatrix}$.
Ⓒ $\begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$. Ⓓ $\begin{pmatrix} -1 & -1.5 \\ 1 & -1 \end{pmatrix}$.
Ⓔ $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

17. (L.O.1)

Find the matrix of the composite transformation $f \circ g$.

Ⓐ $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Ⓑ All other answers are wrong. Ⓒ $\begin{pmatrix} -0.5 & 1.5 \\ 2.5 & 0.5 \end{pmatrix}$. Ⓓ $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$. Ⓔ $\begin{pmatrix} -0.5 & 0.5 \\ -0.5 & 0.5 \end{pmatrix}$.

18. (L.O.1)

Let $E = \{(1,-6)^T, (-3,1)^T\}$ and $F = \{(5,-2)^T, (-9,4)^T\}$ be two bases of $\mathbb{R}^2$ and $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation defined by $T(x,y) = (-x, 3x-2y)$. Find the matrix A_{EF} of T with respect to the bases E and F.

Ⓐ $\begin{pmatrix} 65.5 & -43.5 \\ 36.5 & -24.5 \end{pmatrix}$ Ⓑ $\begin{pmatrix} 20 & -33 \\ -6 & 10 \end{pmatrix}$ Ⓒ $\begin{pmatrix} 52.5 & -93.5 \\ -24.5 & 44 \end{pmatrix}$ Ⓓ $\begin{pmatrix} 22 & 81 \\ -6 & -22 \end{pmatrix}$ Ⓔ Other answers are wrong.

19. (L.O.1) Let $A = \begin{pmatrix} 2 & -5 \\ -12 & -15 \end{pmatrix}$. Which of the following vectors in an eigenvector of A?

Ⓐ $(-3,5)$ Ⓑ $(-5,3)$ Ⓒ All other answers are wrong. Ⓓ $(-3,4)$ Ⓔ $(-5,5)$

Questions 20 through 22 Suppose that $u = (3,1)^T$ is an eigenvector of the matrix A corresponding to $\lambda = -1$, and $v = (5,4)^T$ is an eigenvector of A corresponding to $\lambda = -2$.

20. (L.O.1) Find $det(A)$.

Ⓐ All other answers are wrong. Ⓑ 4 Ⓒ 2 Ⓓ -3 Ⓔ 1

21. (L.O.1) Find the vector $A \times (u - 3v)$.

Ⓐ (21, 26) Ⓑ (27, 23) Ⓒ (26, 21) Ⓓ (23, 27) Ⓔ All other answers are wrong.

22. (L.O.1) Find the $(1,2)$ entry of the matrix A.

Ⓐ -4.1429 Ⓑ 1.8571 Ⓒ All other answers are wrong. Ⓓ -1.6429 Ⓔ -2.1429

23. (L.O.1) Let $f : \mathbb{R}^n \to \mathbb{R}^n$ and $g : \mathbb{R}^n \to \mathbb{R}^n$ be two linear transformations. Choose the WRONG statement.

Ⓐ All other answers are wrong. Ⓑ The composite transformation $g \circ f(x) := g(f(x))$ is a linear transformation. Ⓒ $h(x) := f(x) + g(x)$ is a linear transformation. Ⓓ For all

$\alpha, \beta \in \mathbb{R}$: $h(x) := \alpha f(x) + \beta g(x)$ is a linear transformation. Ⓔ If $n = 1$, then the function $f^g(x) := (f(x))^{g(x)}$ is a linear transformation

Question 24 through 25:

A company took out a loan of 749 (thoudsand dollars) to launch a business project. The loan was divided into three parts, with some borrowed at 8(%), some at 10(%), and some at 12(%). To pay the annual interest of 64.4 (thoundsand dollars), the business needs to determine how much was borrowed at each rate. Additionally, it's known that the amount borrowed at 10(%) is four times the amount borrowed at 12(%).

24. (L.O.1) Find the amount that the company borrowed at the rate of 8 (%) Ⓐ 562.3333 Ⓑ 562.9933 Ⓒ All other answers are wrong. Ⓓ 562.8533 Ⓔ 563.3333

25. (L.O.1) Find the amount that the company borrowed at the rate of 10 (%) Ⓐ 150.3333 Ⓑ 149.3333 Ⓒ All other answers are wrong. Ⓓ 149.9933 Ⓔ 149.8533

Part II: Essay (2 points, 20 minutes)

26. (L.O.1) The rescue force in a densely populated city in the United States consisted of 500 vehicles distributed in three areas: the airport, the East office, and the West office. Each day after the rescue vehicle completes its mission, it will return to one of the three locations above, the closest place to it. According to long-term observation, after each day 15% of rescue force that start from the airport will return to the East office and 5% will return to the West office; Of the force that start from the East office, 5% will return to the Airport and 5% will return to the West office; Of the force that start from the West office, 10% will return to the airport and 5% will return to the East office.

    a. What is the transition matrix? (The 3 states should be in this order: Airport, East Office, West Office)

    b. What are the eigenvalues and corresponding eigenvectors of the transition matrix?

    c. Find the stable distribution vector (the distribution vector that won't change after every day).

    d. Given that today there are $200, 200, 100$ vehicles start from the airport, East Office, West Office, respectively. Using the diagonalization method to find the number of vehicles in each class after 100 days.

Answer for Part II

a. $A = \begin{pmatrix} 0.8 & 0.05 & 0.1 \\ 0.15 & 0.9 & 0.05 \\ 0.05 & 0.05 & 0.85 \end{pmatrix}$

b. $\lambda_1 = 1, E_1 = span\{(1,2,1)^T\}$

$\lambda_2 = 0.75 E_{0.75} = span\{(-1,1,0)^T\}$

$\lambda_3 = 0.8 E_{0.8} = span\{(1,-2,1)^T\}$

c. $Ax_{st} = x_{st} \Rightarrow x_{st}$ is an eigenvector of $\lambda = 1 \Rightarrow x_{st} = a.(1,2,1)^T$.

The stable distribution vector: $(500/4, 500/2, 500/4)^T = (125, 250, 125)^T$.

d. $x_0 = \begin{pmatrix} 200 \\ 200 \\ 100 \end{pmatrix}$

After 100 days: $x_{100} = P.D^{100}P^{-1}.x_0 = \begin{pmatrix} 125 \\ 250 \\ 125 \end{pmatrix}$.

Answers Sheet

Question sheet code 1121:

1 E. 2 A. 3 B. 4 E. 5 E. 6 A. 7 D. 8 B. 9 B. 10 E. 11 A. 12 C. 13 D. 14 D. 15 C. 16 B. 17 C. 18 C. 19 D. 20 B. 21 D. 22 B. 23 D. 24 B. 25 E.

Question sheet code 1122:

1 D. 2 A. 3 D. 4 E. 5 A. 6 A. 7 B. 8 C. 9 C. 10 C. 11 C. 12 C. 13 A. 14 B. 15 B. 16 E. 17 B. 18 D. 19 B. 20 C. 21 C. 22 E. 23 B. 24 C. 25 E.

Question sheet code 1123:

1 B. 2 B. 3 B. 4 A. 5 B. 6 D. 7 C. 8 C. 9 B. 10 E. 11 E. 12 D. 13 E. 14 E. 15 A. 16 E. 17 B. 18 B. 19 D. 20 D. 21 A. 22 E. 23 B. 24 A. 25 D.

Question sheet code 1124:

1 D. 2 C. 3 D. 4 A. 5 E. 6 B. 7 E. 8 C. 9 A. 10 D. 11 E. 12 D. 13 E. 14 A. 15 B. 16 E. 17 E. 18 A. 19 B. 20 C. 21 B. 22 E. 23 E. 24 A. 25 B.

1 1 2 1

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Linear Algebra

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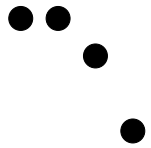
1 1 2 3

MT1007

Linear Algebra

OISP

1 1 2 4



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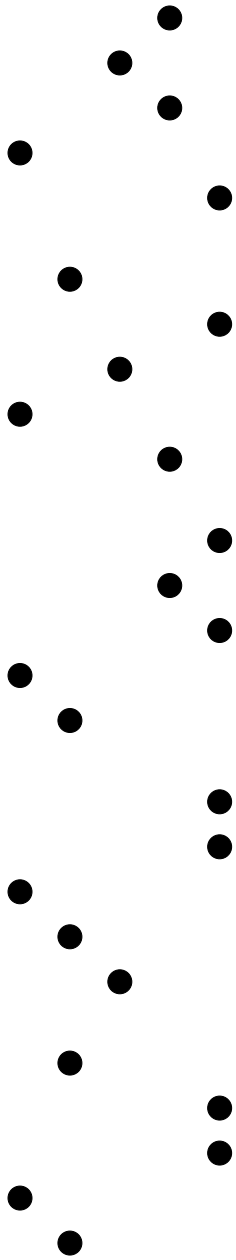
Linear Algebra

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